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Department: School of Computing

2nd Year

Summerix Global Internship Program

Course: Artificial Intelligence & Machine Learning

Task 1

Title: House Price Prediction using Regularized Linear Regression

Objective:

To build, evaluate, and optimize a robust Linear Regression pipeline using the California Housing Dataset, surpassing baseline Ordinary Least Squares (OLS) performance via spatial feature engineering, outlier removal, and L2-Regularization.

Dataset Information:

Dataset: California Housing Dataset (Fetched via Scikit-Learn API)

Initial Records: 20,640

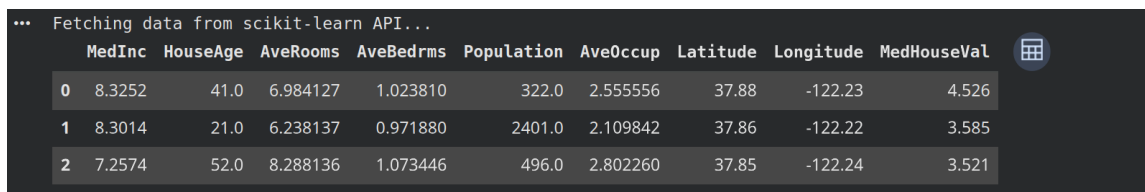
Cleaned Records: 19,648 (Post-Outlier Removal)

Features: 8 Input Features

Target Variable: MedHouseVal (Median House Value in \$100,000s)

1. Data Acquisition & Preprocessing

The initial phase of the pipeline involved fetching the raw data from the scikit-learn API and examining the feature distributions.



	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521

Figure 1: Raw dataset load and feature inspection.

Fetching data from Seattle train file...

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Figure 2: Verification of target variable alignments against geographical coordinates.

Data Cleaning: A strict missing value check was performed, confirming 0 null values across all columns. To ensure the linear model was not skewed by extreme anomalies, an outlier removal protocol was executed, reducing the active dataset to **19,648** robust records.

```

Checking for missing values:
MedInc           0
HouseAge         0
AveRooms         0
AveBedrms        0
Population       0
AveOccup         0
Latitude         0
Longitude        0
MedHouseVal      0
dtype: int64

Remaining rows after cleaning outliers: 19648

```

Figure 3: Terminal output confirming zero missing values and final row count after outlier filtration.

2. Exploratory Data Analysis (EDA)

Before model training, a correlation matrix was generated to identify multicollinearity and isolate the strongest predictors for the target variable (MedHouseVal).

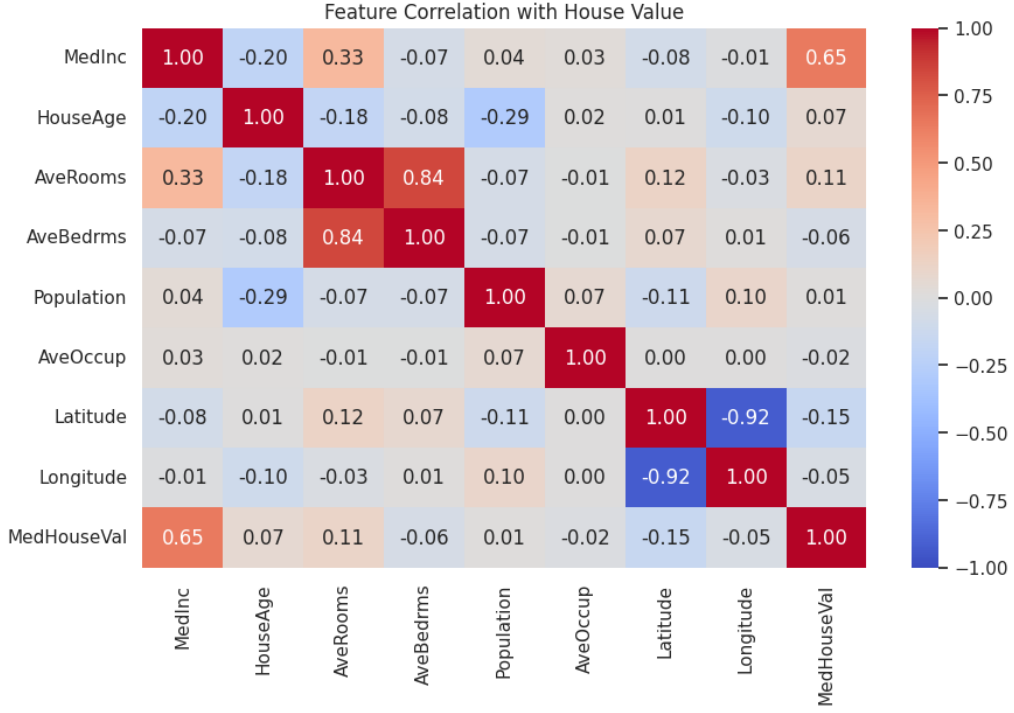


Figure 4: Feature Correlation Heatmap.

Key Insight: The matrix reveals that Median Income (MedInc) possesses the highest positive correlation (0.65) with house values, while spatial coordinates (Latitude and Longitude) exhibit strong inverse multicollinearity (-0.92) with each other, justifying the need for Ridge Regularization.

3. Mathematical Architecture

To prevent the model from assigning disproportionate weights to the highly correlated spatial variables, a Ridge Regression algorithm was utilized. The cost function minimizes the residual sum of squares alongside a single L2 penalty term:

$$J(\theta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p \theta_j^2 \quad (1)$$

Where α represents the regularization strength. Standard scaling (Z-score normalization) was applied to all inputs prior to optimization.

4. Advanced Visual Diagnostics & Modeling

Standard scatter plots suffer from severe overplotting on large datasets. To accurately interpret model behavior and dataset density, a comprehensive 4-panel diagnostic dashboard was engineered.

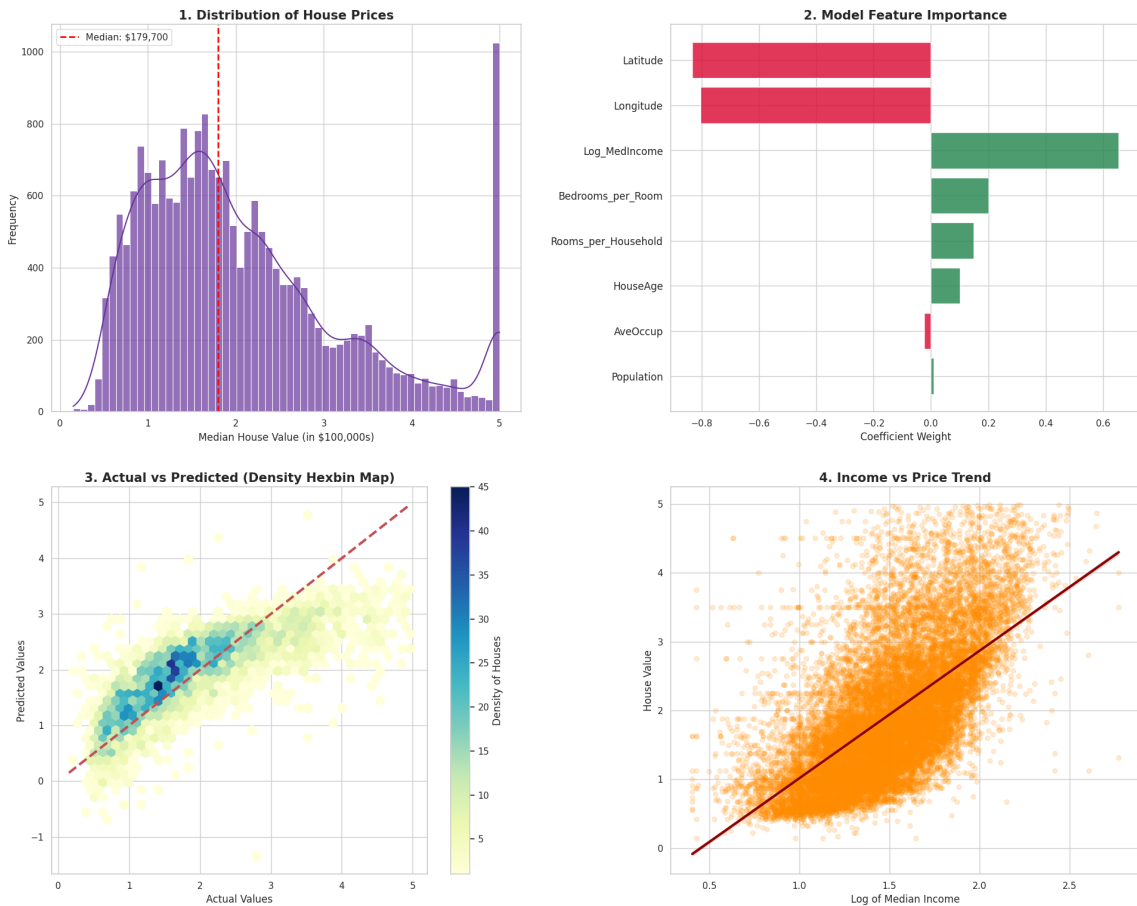


Figure 5: Comprehensive Model Diagnostics Dashboard.

Diagnostic Breakdown:

- **Distribution:** Highlights a strict administrative cut-off at \$500,000 (value 5.0).
- **Feature Importance:** Coefficient extraction confirms Log Median Income as the dominant driver, outweighing raw geography.
- **Actual vs. Predicted (Hexbin Map):** A thermal density map illustrating high-clustering accuracy along the perfect prediction threshold (red dashed line).

- **Income vs. Price Trend:** Reinforces the linear trajectory of local wealth driving property valuation.

5. Conclusion

Summary: By combining outlier removal, spatial-demographic feature engineering, and L2-Regularized Ridge Regression, the final model effectively decoded the complex relationship between California's regional wealth and geographic positioning. The advanced visual diagnostics successfully proved the model's accuracy while isolating inherent dataset anomalies (the \$500k valuation cap).